Technical Specification: Wildlife Camera Trigger & Recording Logic

1. System Parameters (Configurable)

The implementation must allow these variables to be defined in a configuration file:

Parameter	Type	Description
T_VALID	Float (ms)	Minimum high-signal duration to count as a valid pulse.
T_INSTANT	Float (ms)	Minimum high-signal duration to trigger an immediate recording.
N_THRESHOLD	Integer	Number of valid pulses required within T_WINDOW to trigger recording.
T_WINDOW	Float (sec)	Duration of the sliding window for pulse accumulation.
T_PRE_RECORD	Float (sec)	Buffer length to be captured before the trigger event.
T_AFTER_RECORD	Float (sec)	Minimum time the recording continues after the last trigger event.

2. Input Handling (PIR Sensor)

- **Method:** Use **GPIO Edge Detection** (Interrupt-driven).
- **Logic:**
  - Capture timestamps for RISING and FALLING edges to calculate pulse duration P.
  - A pulse is **Valid** if $P \ge T_{VALID}$.
  - A pulse is **Instant-Trigger** if $P \ge T_{INSTANT}$.

3. Core Logic & State Machine

3.1 Trigger Conditions

Recording starts if:

1. **Condition A:** An Instant-Trigger pulse is detected.
2. **Condition B:** The number of Valid pulses within the current sliding window T_WINDOW reaches N_THRESHOLD.

3.2 Recording & Pre-Record

- **Circular Buffer:** The system must maintain a continuous RAM-based circular buffer (length = T_PRE_RECORD).
- **File Management:** Upon trigger, the buffer is flushed to the storage media, and the stream continues.
- **Extension Logic:** If a Valid pulse occurs during an active recording, the recording timer is **reset** (not added) to T_AFTER_RECORD.
- **Safety Constraint:** The recording must not stop as long as the current T_WINDOW (triggered by the latest pulse) is still active.

3.3 IR-LED Control

- **ON:** Immediately when a Valid pulse is detected.
- **OFF:** The LED remains active until **both** conditions are met:
    1. The T_WINDOW timer has expired (no new pulses).
    2. No recording is currently in progress.
- **Override:** The LED must shut off immediately when the recording process terminates.

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4. Implementation Directives for Claude Opus

A. Sliding Window Implementation

Do not use a fixed-interval timer. Use a **List/Queue of timestamps** for valid pulses. On every new pulse:

1. Add the new timestamp to the list.
2. Remove all timestamps older than now() - T_WINDOW.
3. Check if list.length >= N_THRESHOLD.

B. Memory Management (Pre-Record)

To prevent SD card wear-out, the pre-record buffer **must** be kept in a tmpfs (RAM disk) or a memory-mapped byte array. Only commit to disk once a trigger is confirmed.

C. Timer Reset Logic

Ensure the recording end-time is updated using a "Highest Wins" approach:

$$End\_Time = \max(now() + T_{AFTER\_RECORD}, \text{Current\_Window\_Expiry})$$

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5. Critical Constraints (Review Checklist)

- **Race Conditions:** Ensure the GPIO interrupt handler does not conflict with the file-writing thread.
- **Edge Case:** If T_INSTANT is triggered while the system is still evaluating a sequence for N_THRESHOLD, T_INSTANT takes precedence and starts the recording immediately.
- **Debouncing:** Implement a software lockout for 50ms after a FALLING edge to prevent mechanical PIR noise from generating ghost pulses.